

Shoab Ahamed

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🌐 [Kaggle](#) 🐙 [Github](#) 🔗 [LinkedIn](#) 📍 Khulna, BD

Summary

Final Year CS student at KUET who learns how systems work under the hood by building them from the ground up. Systematically studied machine learning daily in the first two years of my university achieving Kaggle Expert rank in Notebooks & Discussion. The same followed into full-stack development — curiosity about how image editing platforms work led me to build ImageCraft — a full-stack image editor with custom WebGL shaders, PyTorch-based ML features, Stripe subscriptions, and Google OAuth.

Education

B.Sc. in Computer Science and Engineering

Feb. 2020 – Present (Final Semester)

Khulna University of Engineering & Technology (KUET)

CGPA: 3.16/4.0

Achievements & Certifications

Kaggle Expert – Notebooks & Discussions

Notebooks covering deep learning, computer vision, and data analysis

Deep Learning Specialization

DeepLearning.AI

[Credential](#)

Google Data Analytics Professional Certificate

Google

[Credential](#)

Technical Skills

Programming Languages

C, C++, Python, JavaScript, PHP

Frameworks & Libraries

React, Flask, Laravel, PyTorch, NumPy, OpenCV, scikit-learn

Graphics & Rendering

OpenGL, WebGL, Fabric.js

Databases

MySQL, MongoDB, Firebase, Oracle

Cloud & Services

Stripe, Cloudinary, Google OAuth

Major Projects

ImageCraft – Full-Stack Image Editing Platform

[Link](#)

- Built a full-stack image editing web app using React (Vite), Fabric.js, and a Flask REST API backed by MongoDB; deployed frontend on Vercel and backend on Render with Cloudinary for production image storage
- Authored 20+ custom WebGL fragment shaders (GLSL) for real-time GPU-accelerated image filters including Gaussian blur, bilateral filter, Sobel/Canny edge detection, swirl, and histogram equalization
- Integrated AdaIN-based neural style transfer and MDSR super-resolution models using PyTorch, serving inference via Flask endpoints with per-user GPU quota tracking tied to subscription tier
- Implemented JWT authentication, Google OAuth 2.0, and Stripe subscriptions with webhook handlers for payment events, plan upgrades, cancellations, and automated quota resets.
- Built a community gallery with search, pagination, rating, bookmarking, and view tracking alongside a content reporting system and admin dashboard

3D Mosque — Real-Time OpenGL Simulation

[Link](#)

- Built an interactive real-time 3D scene in C++ using Modern OpenGL 3.3, constructing all geometry from hand-coded primitives, fractals, cubic bezier
- Wrote GLSL shaders with 36 point lights, 1 directional, and 1 spot light
- Implemented gravity-driven particle system for animated water flow and controllable animations for doors, windows, fans, and bookshelves, lightning

University Management System

[Link](#)

- Built a role-based university management web app using Laravel, MySQL, and Blade with custom middleware enforcing admin and student access controls
- Implemented full CRUD for 6 entities (notices, events, teachers, courses, halls, students) with server-side validation, image uploads, and an admin dashboard

TLS Protocol Simulation

[Link](#)

- Implemented a basic TLS protocol simulation using the OMNeT++ framework with ECDHE from scratch
- Demonstrated limitations in the absence of certificate-based authentication, including vulnerability to man-in-the-middle attacks